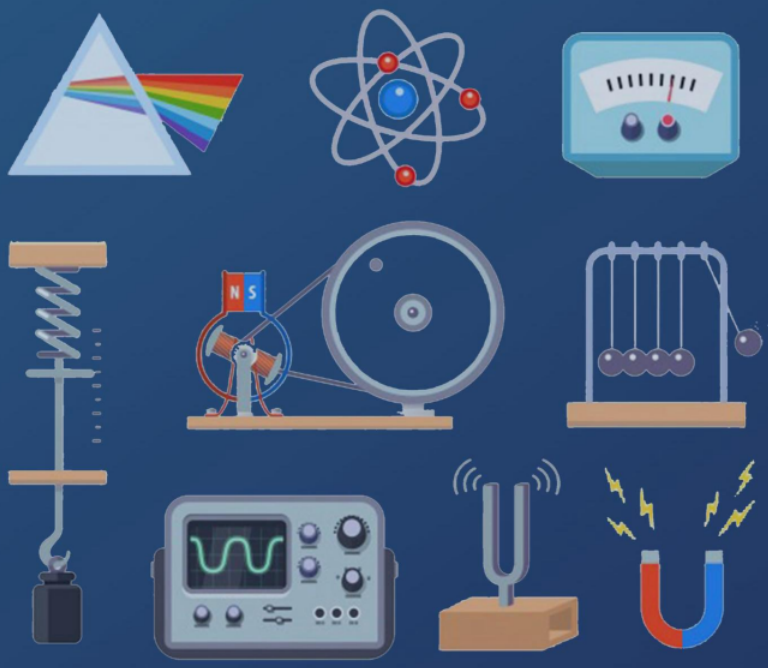
PHYSICS

# Electrostatics




SHEET

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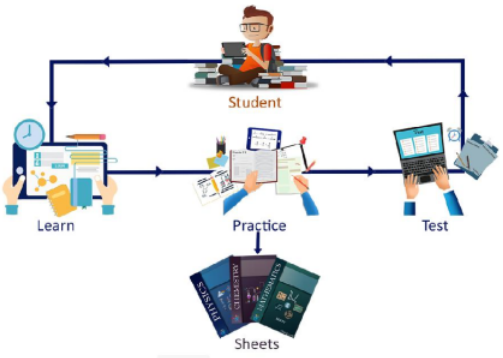
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***Note***

*Detailed solutions are available on the eSaral App.*

# *Module Description*




For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

*"Practice makes a man perfect"*

**1. Concept builder – 1 & 1A**

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

**2. Brain Booster – 2 & 2A**

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

**3. Simulator – JM & JA**

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

# *Home Work Index*

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

| Sr.No. | Topic Name | Ex 1 | Ex 1A | Ex 2 | Ex 2A |
|---|---|---|---|---|---|
| 1 | Electric Charge | 1 | | | |
| 2 | Coulomb's Law | 2-4 | | 1,2 | 16-18 |
| 3 | Charging by Friction & Induction | | 23 | | |
| 4 | Electric Field | 5,6 | 1,2 | 3 | |
| 5 | EFI on Axis of Uniformly Charged Ring | | | 4 | 1 |
| 6 | EFI Due to Thin Wire | 7 | | | |
| 7 | EFI due to Thin Circular Arc | | | 5 | |
| 8 | Electric Field Lines | 8-10 | | | |
| 9 | Electric Flux | 11,12 | 22 | | 19,20 |
| 10 | Gauss Law | 13 | 3-5 | 6,7 | 2 |
| 11 | Gauss Law Application - EFI due to Sphere and Thin Shell | 14 | | | |
| 12 | Gauss Law Application - EFI due to Sphere | 15,16 | | 8,9 | 3 |
| 13 | Gauss Law Application - EFI due to Solid and Hollow Cylinder , Infinite Sheet | 17 | 6,24 | | 4 |
| 14 | Electrostatic Potential Energy | 18 | | | |
| 15 | Applying Work Energy Theorem | 19,20 | 7 | 10,11,12 | 5,6,21-23 |
| 16 | Electric Potential | 21 | | | |
| 17 | Calculation of Electric Potential | 22 | 8,9,19,20 | | |
| 18 | Calculating Potential from Field | 23-25 | 10,11,12,13 | | 7 |
| 19 | Calculating Field from Potential | 26 | 25 | | 8 |
| 20 | Potential due to Sphere and Thin Shell | 27 | | 13,14 | 9 |
| 21 | Equipotential Surface | 28 | 14 | | |
| 22 | Electric Dipole and Dipole Moment | 29 | | | |
| 23 | Electris Field and Potential due to Dipole | 30 | | 15-17 | |
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| 26 | Conductors and its Properties | 32 | | | |
| 27 | EFI on Surface of Conductors | 33 | | | |
| 28 | Electric Potential due to Conductors | 34,35 | | 18-20 | |
| 29 | Conductors with Cavity | 36-38 | | | 24 |
| 30 | Parallel Conducting Plates | | 17 | | |
| 31 | Electrostatic Pressure, Energy Density and Effect of Medium | | | | |
| 32 | After full Chapter | 39 | 18 | | 11-15,25 |

## Exercise - 1 Concept Builder

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

### Electric Charge

**Q1.** Five balls numbered 1 to 5 are suspended using separate threads. Pairs (1,2); (2,4) and (4,1) show electrostatic attraction, while pairs (2,3) and (4,5) show repulson. The ball 1 may be :

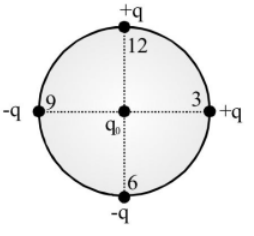
(A) positively charged
(B) negatively charged
(C) either (+ve) or (–ve)
(D) neutral

### Coulomb's Law

**Q2.** Two equal and like charges when placed 5 cm apart experience a repulsive force of 0.144 newton. The magnitude of the charge in micro–coulomb will be

(A) 0.2 (B) 2
(C) 20 (D) 12

**Q3.** Four charges are placed at the circumference of dial clock as shown in figure. If the clock has only hour hand, then the resultant force on a charge $q_0$ placed at the centre, points in the direction which show the time as :-



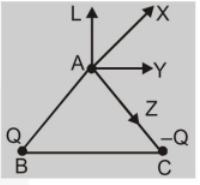

(A) 1:30 (B) 7:30
(C) 4:30 (D) 10:30

**Q4.** The dimensions of $\in_r$ is :–

(A) $[M^{-1} L^{-3} T^4 A^2]$ (B) $[M^0 L^{-3} T^2 A^3]$
(C) $[M^{-1} L^{-3} T^3 A]$ (D) Dimensionless

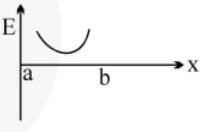
### Electric Field

**Q5.** For the given figure the direction of electric field at A will be :




(A) towards AL (B) towards AY
(C) towards AX (D) towards AZ

**Q6.** Two point like charges a & b whose magnitudes are same are positioned at a certain distance from each other, a is at origin. Graph is drawn between electric field strength and distance x from a. E is taken positive if it is along the line joining from a to b




(A) a is positive, b is negative
(B) a & b both are positive
(C) a & b both are negative
(D) a is negative, b is positive

### EFI Due to Thin Wire

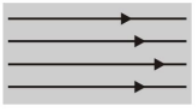
**Q7.** Two infinitely long parallel wires having linear charge densities $\lambda_1$ and $\lambda_2$ respectively are placed at a distance of R. The force per unit length of an either wire will be :–

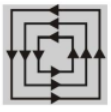
(A) $\frac{2k\lambda_1\lambda_2}{R^2}$ (B) $\frac{2k\lambda_1\lambda_2}{R}$

(C) $\frac{k\lambda_1\lambda_2}{R^2}$ (D) $\frac{\lambda_1\lambda_2}{R}$

## Electric Field Lines

**Q8.** Which one of the following pattern of electrostatic line of force can't possible :–

(A)  (B) 

(C)  (D) 

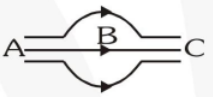
**Q9.** The figure shows some of the electric field lines corresponding to an eletric field. The figure suggests

(A) $E_A > E_B > E_C$

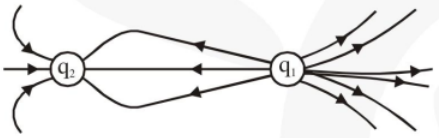
(B) $E_A = E_B = E_C$

(C) $E_A = E_C > E_B$

(D) $E_A = E_C < E_B$



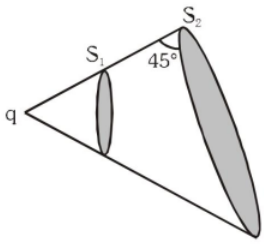

**Q10.** For the figure shown, what is the ratio of the charges $q_2/q_1$, where the figure shown has a representation of the field lines in the space near the charges.




(A) –3/2 (B) –2/3 (C) 2/3 (D) 3/2

## Electric Flux

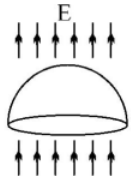
**Q11.** In the given figure flux through surface $S_1$ is $\phi_1$ & through $S_2$ is $\phi_2$. Which is correct?




(A) $\phi_1 = \phi_2$ (B) $\phi_1 > \phi_2$

(C) $\phi_1 < \phi_2$ (D) None of these

**Q12.** A hemispherical surface (half of a spherical surface) of radius R is located in a uniform electric field E that is parallel to the axis of the hemisphere. What is the magnitude of the electric flux through the hemisphere surface?

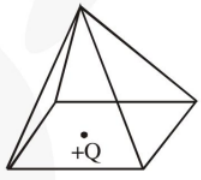



(A) 0 (B) $4\pi R^2E/3$

(C) $2\pi R^2E$ (D) $\pi R^2E$

## Gauss Law

**Q13.** A point charge +Q is positioned at the center of the base of a square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is




(A) $\frac{Q}{16\varepsilon_0}$ (B) $\frac{Q}{4\varepsilon_0}$

(C) $\frac{Q}{8\varepsilon_0}$ (D) None of these

## Gauss Law Application - EFI due to Sphere and Thin Shell

**Q14.** Electric field inside a hollow uniformly charged sphere is :–

(A) Zero

(B) proportional to a distance from centre

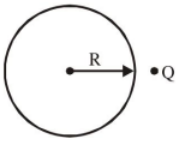
(C) inversly proportional to the distance

(D) inversly proportional to square of distance

## Gauss Law Application - EFI due to Sphere

**Q15.** **Statement 1:** A charge is outside the Gaussian sphere of radius R. Then electric field on the surface of sphere is zero.

**Statement 2** : As $\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\varepsilon_0}$, for the sphere $q_{in}$ is zero, so $\oint \vec{E} \cdot d\vec{s} = 0$.



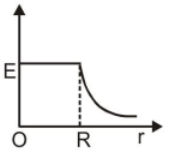

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

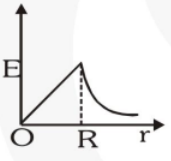
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

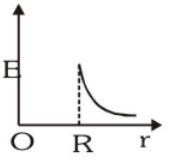
(C) Statement-1 is true, statement-2 is false.

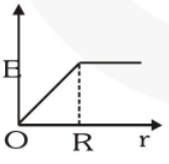
(D) Statement-1 is false, statement-2 is true.

**Q16.** The electric field due to a uniformly charged non conducting sphere of radius R as a function of the distance from its centre is represented graphically by

(A) 

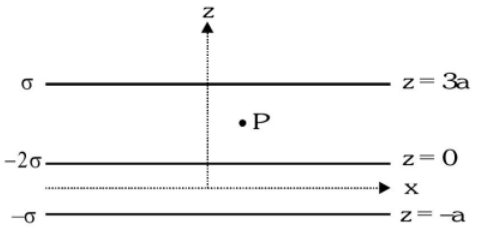

(B) 


(C) 


(D) 


## Gauss Law Application - EFI due to Solid and Hollow Cylinder, Infinite Sheet

**Q17.** Three infinitely long charge sheets are placed which charge density as shown in figure. The electric field at point P is :–

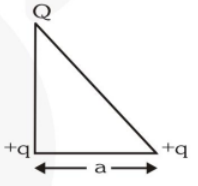



(A) $\frac{2\sigma}{\varepsilon_0}\hat{k}$ (B) $-\frac{2\sigma}{\varepsilon_0}\hat{k}$

(C) $\frac{4\sigma}{\varepsilon_0}\hat{k}$ (D) $-\frac{4\sigma}{\varepsilon_0}\hat{k}$

## Electrostatic Potential Energy

**Q18.** Three charges Q, +q and +q are placed at the vertices of a right angle triangle (isoscles triangle) as shown. The net electrostatic energy of the configuration is zero, if Q is equal to :

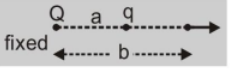



(A) $\frac{-q}{1+\sqrt{2}}$ (B) $\frac{-2q}{2+\sqrt{2}}$

(C) –2q (D) +q

## Applying Work Energy Theorem

**Q19.** A charge q = $10^{-6}$ C of mass 2 g (fig.) is free to move then calculate its speed, when it is at a distance of b.[Assume a = 1 m,b = 10 m, Q = $10^{-3}$ C]




(A) 90 m/s. (B) 9 m/s.

(C) 900 m/s. (D) none of these

**Q20.** A particle of mass m and charge q is released from rest in an electric field E. Then the K. E after time t will be:

(A) $\frac{2E^2t^2}{mq}$ (B) $\frac{E^2q^2t^2}{2m}$

(C) $\frac{Eq^2m}{2t^2}$ (D) $\frac{Eqm}{2t}$

## Electric Potential

**Q21.** At any point on the perpendicular bisector of line joining two equal and opposite charges

(A) The electric field is zero

(B) The electric potential is zero

(C) The electric potential decreases with increasing distance from centre

(D) The electric field is perpendicular to the line joining the charges

## Calculation of Electric Potential

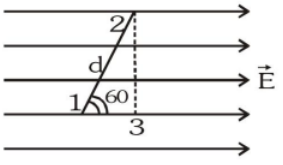
**Q22.** The electric potential and electric field at any given position due to a point charge are 600 V and 200 N/C respectively. Then magnitude of point charge would be :

(A) 3 μC (B) 30 μC

(C) 0.2 μC (D) 0.5 μC

## Calculating Potential from Field

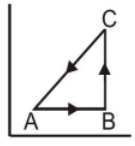
**Q23.** The electric field $\vec{E}$ between two points is constant in both magnitude and direction. Consider a path of lenght d at an angle $\theta = 60^0$ with respect to field lines shown in figure. The potential difference between points 1 and 2 is :




(A) $\frac{E}{d\cos 60°}$ (B) $Ed\cos 60°$

(C) $\frac{Ed}{\cos 60°}$ (D) $\frac{E}{d}\cos 60°$

**Q24.** As shown in figure, on bringing a charge Q from point A to B and from B to C, the work done are 2 joule and – 3 joule respectively. The work done in bringing the charge from C to A will be



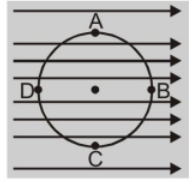

(A) – 1 joule

(B) 1 joule

(C) 2 joule

(D) 5 joule

**Q25.** A circle of radius R is drawn in a uniform electric field E as shown in the fig. $V_A$, $V_B$, $V_C$ and $V_D$ are respectively the potential of point A, B, C and D at the periphery of the circle then -




(A) $V_A > V_C$, $V_B = V_D$

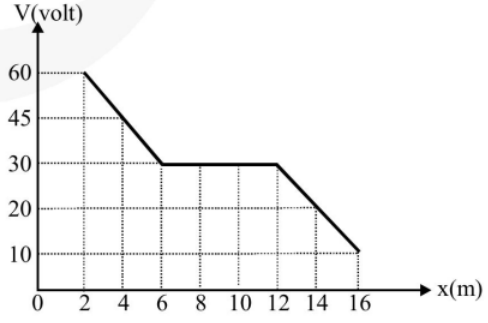
(B) $V_A < V_C$, $V_B = V_D$

(C) $V_A = V_C$, $V_B < V_D$

(D) $V_A = V_C$, $V_B > V_D$

## Calculating Field from Potential

**Q26.** The variation of potential with distance x from a fixed point is shown in figure. The electric field at x =13m is -

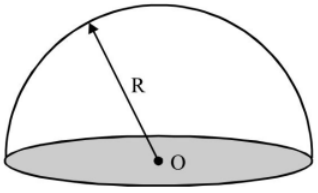



(A) 7.5 volt/meter (B) –7.5 volt/meter

(C) 5 volt/meter (D) –5 volt/meter

## Potential due to Sphere and Thin Shell

**Q27.** Charge $Q$ coulombs is uniformly distributed throughout the volume of a solid hemisphere of radius $R$ metres. Then the potential at centre $O$ of the hemisphere in volts is -

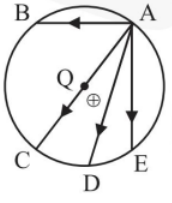



(A) $\frac{1}{4\pi\varepsilon_0}\frac{3Q}{2R}$ (B) $\frac{1}{4\pi\varepsilon_0}\frac{3Q}{4R}$

(C) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{4R}$ (D) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{8R}$

## Equipotential Surface

**Q28.** In the electric field of charge Q, another charge is carried from A to B, A to C, A to D and A to E, then work done will be :-




(A) minimum along path AB
(B) minimum along path AD
(C) minimum along path AE
(D) zero along all the paths

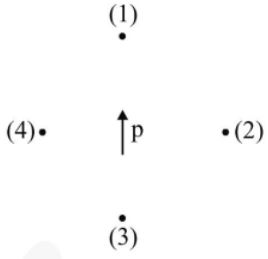
## Electric Dipole and Dipole Moment

**Q29.** Two charges +2C and -2C are kept 3 mm apart. What is the magnitude of dipole moment of this setup.

(A) 0 (B) 6 Cm
(C) 0.006 Cm (D) 2 Cm

## Electric Field and Potential due to Dipole

**Q30.** The drawing shows four points surrounding an electric dipole. Which one of the following expressions best ranks the electric potential at these four locations?




(A) 1 > 2 = 4 > 3
(B) 3 > 2 > 4 > 1
(C) 3 > 2 = 4 > 1
(D) 2 = 4 > 1 = 3

## Dipole in External Electric Field

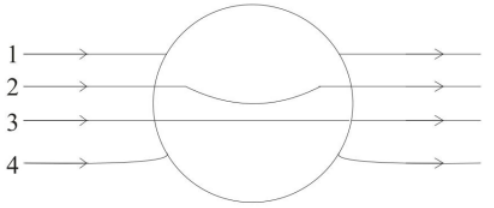
**Q31.** A point particle of mass M is attached to one end of a massless rigid non-conducting rod of length L. Another point particle of same mass is attached to the other end of the rod. The two particles carry charges +q and – q repectively. This arrangement is held in a region of uniform electric field E such that the rod makes a small angle $\theta$ ($< 5^0$) with the field direction. The minimum time needed for the rod to become parallel to the field after it is set free. (rod rotates about centre of mass)

(A) $2\pi\sqrt{\frac{ML}{2qE}}$ (B) $\pi\sqrt{\frac{ML}{2qE}}$

(C) $\frac{\pi}{2}\sqrt{\frac{ML}{2qE}}$ (D) $4\pi\sqrt{\frac{ML}{2qE}}$

## Conductors and its Properties

**Q32.** A metallic solid sphere is placed in a uniform electric field. The lines of force follow the path (s) shown in figure as :

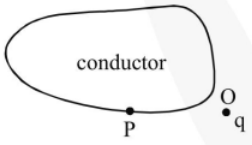



(A) 1 (B) 2 (C) 3 (D) 4

## EFI on Surface of Conductors

**Q33.** The density of charge at P on the conductor is σ. The resultant electric field near P will [$\hat{n}$ = unit normal vector at point P.

$\vec{r}$ = vector along OP] where K = 1/4 $\pi\varepsilon_0$




(A) $\left(\frac{\sigma}{\epsilon_0}\right)\hat{n}+\left(\frac{Kq\vec{r}}{r^3}\right)$

(B) $\left(\frac{\sigma}{2\epsilon_0}\right)\hat{n}+\left(\frac{Kq\vec{r}}{r^3}\right)$

(C) $\left(\frac{\sigma}{\epsilon_0}\right)\hat{n}$

(D) $\left(\frac{\sigma}{2\epsilon_0}\right)\hat{n}$

## Electric Potential due to Conductors

**Q34.** **Statement–1 :** If a point charge q is placed in front of an infinite grounded conducting plane surface, the point charge will experience a force.

**Statement–2 :** This force is due to the induced charge on the conducting surface which is at zero potential.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

**Q35.** **Statement–1 :** If two concentric conducting sphere which are connected by a conducting wire. No charge can exist on inner sphere.

**Statement–2 :** When charge on outer sphere will exist then potential of inner shell and outer shell will be same.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is False.

(C) Statement–1 is False, Statement–2 is True.

(D) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.

## Conductors with Cavity

**Q36.** **Statement–1 :** Induced charge does not contribute to electric field or potential at a given point.

**Statement–2 :** A point charge $q_0$ is kept outside a solid metallic sphere, the electric field inside the sphere is zero.

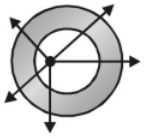
(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

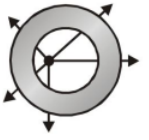
(B) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.

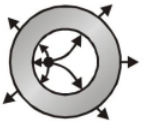
(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True.

**Q37.** A metallic shell has a point charge 'q' kept inside its cavity. Which one of the following diagrams correctly represents the electric lines of forces :-

(A) 

(B) 

(C) 

(D) 

**Q38.** A spherical portion has been removed from a solid sphere having a charge distributed uniformly in its volume in the figure. The electric field inside the emptied space is :-



(A) Zero everywhere

(B) Non-zero and uniform

(C) Non-uniform

(D) Zero only at its center

## Miscellaneous

**Q39.** Under the influence of the electric field of charge +Q, a charge –q is moving around it in an elliptical orbit. Find out the correct statement(s) :-

(A) The angular momentum of the charge –q is constant

(B) The linear momentum of the charge –q is constant

(C) The angular velocity of the charge –q is constant

(D) The linear speed of the charge –q is constant

# *Exercise - 1A* *Concept Builder*

## ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s).* ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** A particle of mass m and charge q is thrown in a region where uniform gravitational field and electric field are present. The path of particle :

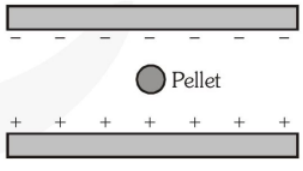
(A) May be a straight line

(B) May be a circle

(C) May be a parabola

(D) May be a hyperbola

**Q2.** A tiny charged pellet of mass *m* is suspended at rest by the electric field between two horizontal, charged metallic plates. The lower plate has a positive charge and the upper plate has a negative charge. Which statement in the answers here is/are true ?




(A) The electric field between the plates points vertically upward.

(B) The pellet is negatively charged.

(C) The magnitude of the electric force on the pellet is equal to *mg*.

(D) If the magnitude of charge on the plates is increased, the pellet begins to move upward.

**Q3.** If the flux of the electric field through a closed surface is zero,

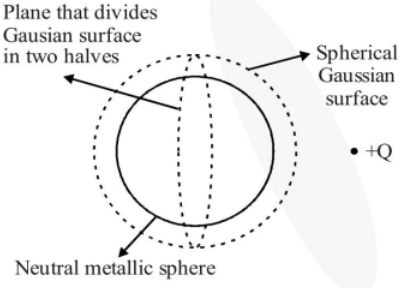
(A) The electric field must be zero everywhere on the surface

(B) The electric field may be zero everywhere on the surface

(C) The net charge inside the surface must be zero

(D) The net charge in the vicinity of the surface must be zero.

**Q4.** Figure shows a neutral metallic sphere with a point charge +Q placed near its surface. Electrostatic equilibrium conditions exist on metallic sphere. Mark the correct statements:




(A) Net flux through Gaussian surface due to charge Q is zero.

(B) Net flux through Gaussian surface due to charges appearing on the outer surface of metallic sphere must be zero.

(C) If point charge Q is displaced towards metallic sphere, magnitude of net flux through right hemispherical closed Gaussian surface increases.

(D) If point charge Q is displaced towards metallic sphere, charge distribution on outer surface of sphere will change.

**Q5.** A hollow closed conductor of irregular shape is given some charge. Which of the following statements are correct ?

(A) The entire charge will appear on its outer surface.

(B) All points on the conductor will have the same potential.

(C) All points on its surface will have the same charge density.

(D) All points just outside it will have the same electric intensity.

**Q6.** In the statement of Gauss's law :

$$\oiint \vec{E}.d\vec{s} = \frac{q_{enclosed}}{\epsilon_0}$$

Select the correct alternative(s) :

(A) The value of $\oiint \vec{E}.d\vec{s}$ does not depend on the shape of gaussian surface if it encloses same charge.

(B) The electric field $\vec{E}$ also includes the contribution of charges outside the gaussian surface.

(C) If charge is moved inside the gaussian surface $\vec{E}.d\vec{s}$ at any point on surface remains constant.

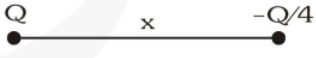
(D) If charge is moved inside the gaussian surface electric field at every point on surface may change but $\oiint \vec{E}.d\vec{s}$ remains unchanged.

**Q7.** A particle of charge 1μC & mass 1 gm moving with a velocity of 4 m/s is subjected to a uniform electric field of magnitude 300 V/m for 10 sec. Then it's final speed cannot be :

(A) 0.5 m/s (B) 4 m/s

(C) 3 m/s (D) 6 m/s

**Q8.** Two point charges Q and –Q/4 are separated by a distance x. Then :




(A) Potential is zero at a point on the axis which is at a distance x/3 on the right side of the charge–Q/4

(B) Potential is zero at a point on the axis which is at a distance x/5 on the left side of the charge–Q/4

(C) Electric field is zero at a point on the axis which is at a distance x on the right side of the charge–Q/4

(D) There exist two points on the axis where electric field is zero

**Q9.** Four charges of 1μC, 2μC, 3μC, and – 6μC are placed one at each corner of the square of side 1m. The square lies in the x–y plane with its centre at the origin.

(A) The electric potential is zero at the origin.

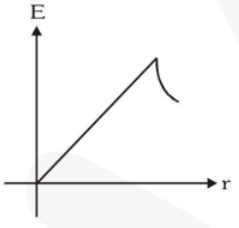
(B) The electric potential is zero everywhere along the x–axis only if the sides of the sqaure are parallel to x and y axis.

(C) The electric potential is zero everywhere along the z–axis for any orientation of the square in the x–y plane.

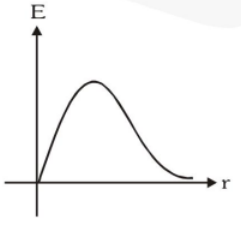
(D) The electric potential is not zero along the z–axis except at the origin.

**Q10.** A circular ring carries a uniformly distributed positive charge. The electric field (E) and potential (V) varies with distance (r) from the centre of the ring along its axis as :

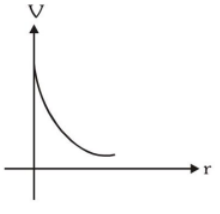
(A)



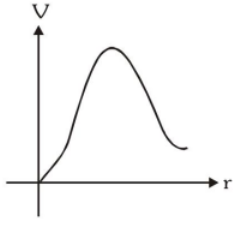

(B)




(C)




(D)




**Q11.** Which of the following is true for the figure showing electric lines of force?

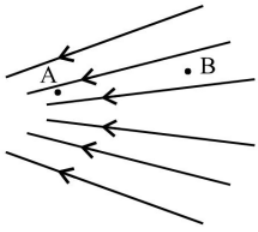
(E is electrical field, V is potential)

(A) $E_A > E_B$

(B) $E_B > E_A$

(C) $V_A > V_B$

(D) $V_B > V_A$




**Q12.** The electric potential decreases uniformly from V to –V along X–axis in a coordinate system as we moves from a point $(-x_0, 0)$ to $(x_0, 0)$, then the electric field at the origin :

(A) must be equal to $\frac{V}{x_0}$

(B) may be equal to $\frac{V}{x_0}$

(C) must be greater than $\frac{V}{x_0}$

(D) may be less than $\frac{V}{x_0}$

**Q13.** A uniform electric field pointing in positive x–direction exists in a region. Let A be the origin, B be the point on the x–axis at x = +1 cm and C be the point on the y–axis at y = +1cm. Then the potentials at the A, B and C satisfy :

(A) $V_A < V_B$ (B) $V_A > V_B$

(C) $V_A < V_C$ (D) $V_A > V_C$

**Q14.** Equipotential surfaces :-

(A) Are closer in regions of large electric fields compared to regions of lower electric fields.

(B) Will be more crowded near sharp edges of a conductor.

(C) Will be more crowded near regions of large charge densities.

(D) Will always be equally spaced.

**Q15.** An electric dipole is placed in an electric field generated by a point charge –

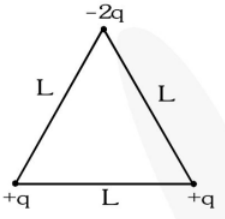
(A) The net electric force on the dipole must be zero.

(B) The net electric force on the dipole may be zero.

(C) The torque on the dipole due to the field must be zero.

(D) The torque on the dipole due to the field may be zero.

**Q16.** Three points charges are placed at the corners of an equilateral triangle of side L as shown in the figure:



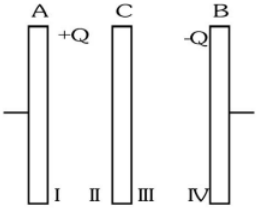

(A) The potential at the centroid of the triangle is zero.

(B) The electric field at the centroid of the triangle is zero.

(C) The dipole moment of the system is $\sqrt{2}$ qL.

(D) The dipole moment of the system is $\sqrt{3}$ qL.

**Q17.** Plates A and B constitutes an isolated, charge parallel plate capacitor. The inner surfaces ( I and IV) of A and B have charges + Q and –Q respectively. A third plate C with charge + Q is now introduced midway between A and B. Which of the following statements is not correct?



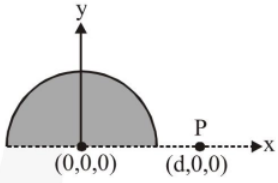

(A) The surface I and II will have equal and opposite charges.

(B) The surfaces III and IV will have equal and opposite charges.

(C) The charge on surace III will be greater than Q.

(D) The potential difference between A and C will be equal to the potential difference between C and B.

**Q18.** A solid hemispherical uniform charged body having charge Q is kept symmetrically along the y-axis as shown in figure. The electric potential at a distance d from the origin along the x-axis at point P will be-




(A) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$

(B) Less than $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$

(C) More than $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$ and less then $\frac{2}{4\pi\varepsilon_0}\frac{Q}{d}$

(D) More than $\frac{2}{4\pi\varepsilon_0}\frac{Q}{d}$

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

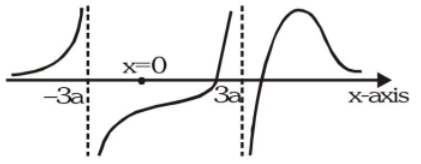
**Paragraph for Question No. 19 & 20**

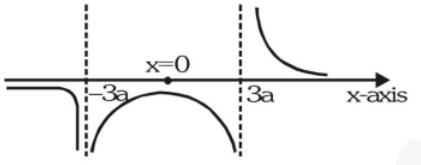
Electric potential is a scalar quantity. Due to a point charge q at distance r, the potential is given by $V = \frac{q}{4\pi\in_0 r}$. A point charge q is placed at (3a, 0) and another charge –2q is placed at (–3a 0).

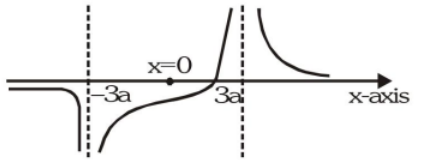
**Q19.** At how many points on the x–axis, (at finite distance) electric potential will be zero ?

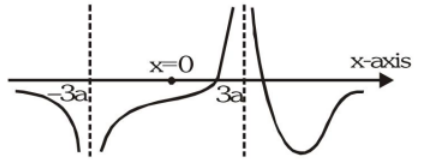
(A) 1 (B) 2 (C) 3 (D) 4

**Q20.** If we plot a graph of potential (V) on x–axis it will be like :

(A)



(B)


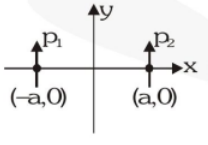
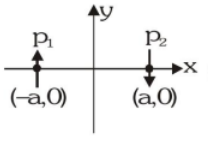
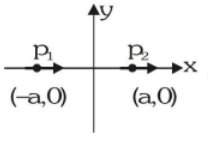
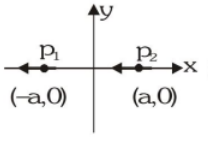

(C)



(D)



## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q21.** In each situation of column–I, two electric dipoles having dipole moments $\vec{p}_1$ and $\vec{p}_2$ of same magnitude (that is, $p_1 = p_2$) are placed on x–axis symmetrically about origin in different orientations as shown. In column–II certain inferences are drawn for these two dipoles. Then match the different orientations of dipole in column–I with the corresponding results in column–II.

| **Column I** | **Column II** |
|---|---|
| (A)   ($\vec{p}_1$ and $\vec{p}_2$ are perpendicular to x–axis as shown) | (P) The torque on one dipole due to other is zero. |
| (B)   ($\vec{p}_1$ and $\vec{p}_2$ are perpendicular to x–axis as shown) | (Q) The potential energy of one dipole in electric field of other dipole is negative |
| (C)   ($\vec{p}_1$ and $\vec{p}_2$ are parallel to x–axis as shown) | (R) There is one straight line in x-y plane (not at infinity) which is equipotential |
| (D)   ($\vec{p}_1$ and $\vec{p}_2$ are parallel to x–axis as shown) | (S) Electric field at origin is zero. |

## NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

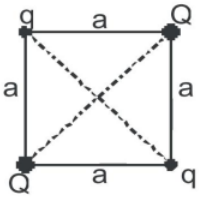
**Q22.** Consider $\vec{E} = 3 \times 10^3 \hat{i}$ (N/C) then what is the flux through the square of 10 cm side, if the normal of its plane makes a 60° angle with the X axis.

## SUBJECTIVE TYPE

**Q23.** When a piece of polythene is rubbed with wool, a charge of $-2 \times 10^{-7}$ C is developed on polythene. What is the amount of mass, which is transfered to polythene.

**Q24.** Find the electric field due to an infinitely long cylindrical charge distribution of radius R and having linear charge density λ at a distance half of the radius from its axis.

**Q25.** For the system shown in figure find Q for which resultant force on q is zero :-

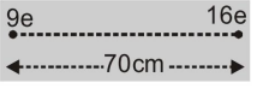



# *Exercise - 2 Brain Booster*

## SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

**Q1.** Two charges are placed as shown in figure. Where should be a third charge be placed so that it remains in rest condition:-




(A) 30cm from 9e (B) 40cm from 16e
(C) 40cm from 9e (D) (A) and (B) Both

**Q2.** A charge Q is divided in two parts $Q_1$ and $Q_2$ and these charges are placed at a distance R. There will be maximum repulsion between them when :-

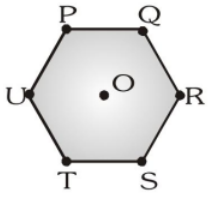
(A) $Q_1 = \frac{Q-Q}{R}, Q_2 = \frac{Q}{R}$

(B) $Q_1 = \frac{2Q}{3}, Q_2 = \frac{Q}{3}$

(C) $Q_1 = \frac{3Q}{4}, Q_2 = \frac{Q}{4}$

(D) $Q_1 = Q_2 = \frac{Q}{2}$

**Q3.** Six charges, three positive and three negative of equal magnitude are to be placed starts from P in clockwise order at the vertices of a regular hexagon such that the electric field at O is double the electric field when only one positive charge of same magnitude is placed at R. Which of the following arrangements of charge is possible for, P, Q, R, S, T and U respectively ?



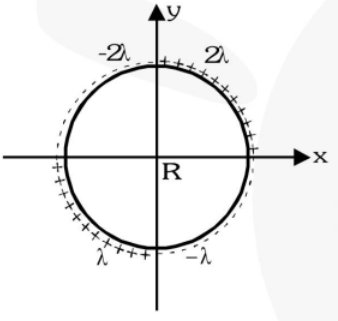

(A) +, –, +, –, –, +  (B) +, –, +, –, +, –
(C) +, +, –, +, –, –  (D) –, +, +, –, +, –

**Q4.** A ring of radius R is charged uniformly with a charge + Q . The electric field at any point on its axis at a distance r from the circumference of the ring will be :-

(A) $\frac{KQ}{r}$  (B) $\frac{KQ}{r^2}$

(C) $\frac{KQ}{r^3}\left(r^2-R^2\right)^{1/2}$  (D) $\frac{KQr}{R^3}$

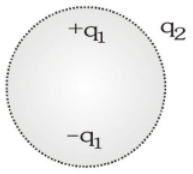
**Q5.** The charge per unit length of the four quadrant of the ring is $2\lambda$, $-2\lambda$, $\lambda$ and $-\lambda$ respectively. The electric field at the centre is :




(A) $\frac{-\lambda}{2\pi\varepsilon_0 R}\hat{i}$  (B) $\frac{\lambda}{2\pi\varepsilon_0 R}\hat{j}$

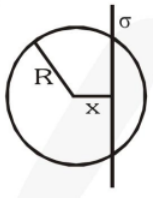
(C) $\frac{\sqrt{2}\lambda}{4\pi\varepsilon_0 R}\hat{i}$  (D) None

**Q6.** Consider the charge configuration and a spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the electric field will be due to:




(A) $q_2$
(B) only the positive charges
(C) all the charges
(D) $+q_1$ and $-q_1$

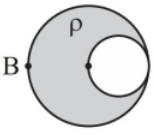
**Q7.** An infinite, uniformly charged sheet with surface charge denisty $\sigma$ cuts through a spherical Gaussian surface of radius R at a distance x from its center, as shown in the figure. The electric flux $\phi$ through the Gaussian surface is :




(A) $\frac{\pi R^2\sigma}{\varepsilon_0}$  (B) $\frac{2\pi\left(R^2-x^2\right)\sigma}{\varepsilon_0}$

(C) $\frac{\pi\left(R-x\right)^2\sigma}{\varepsilon_0}$  (D) $\frac{\pi\left(R^2-x^2\right)\sigma}{\varepsilon_0}$

**Q8.** A positively charged sphere of radius $r_0$ carries a volume charge density $\rho$ (Figure). A spherical cavity of radius $r_0/2$ is then scooped out and left empty, as shown. What is the direction and magnitude of the electric field at point B ?




(A) $\frac{17\rho r_0}{54\,\epsilon_0}$ left  (B) $\frac{\rho r_0}{6\,\epsilon_0}$ left

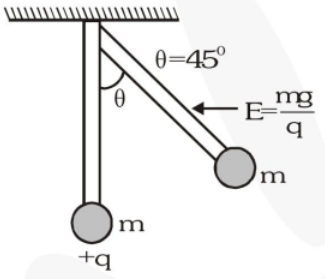
(C) $\frac{17\rho r_0}{54\,\epsilon_0}$ right  (D) $\frac{\rho r_0}{6\,\epsilon_0}$ right

**Q9.** A sphere of radius R carries charge density proportional to the square of the distance from the center: $\rho = Ar^2$, where A is a positive constant. At a distance of R/2 from the center, the magnitude of the electric field is -

(A) $A/4\pi\varepsilon_0$ (B) $AR^3/40\varepsilon_0$

(C) $AR^3/24\varepsilon_0$ (D) $AR^3/5\varepsilon_0$

**Q10.** A horizontal electric field (E = mg/q) exists as shown in figure and a mass m attached at the end of a light rod. If mass m is released from the position shown in figure, find the angular velocity of the rod when it passes through the bottom most position :




(A) $\sqrt{\frac{g}{\ell}}$ (B) $\sqrt{\frac{2g}{\ell}}$

(C) $\sqrt{\frac{3g}{\ell}}$ (D) $\sqrt{\frac{5g}{\ell}}$

**Q11.** In an electrostatic field, a point charge is taken from A to B such that external work done on it is $W_1$ and its kinetic energy increases by $K_1$. In the same field the same charge is taken from A to C such that work done by electrostatic field on it is $W_2$ and its kinetic energy increase by $K_2$. Find the external work done on the charge if it is taken from B to C without changing K.E.

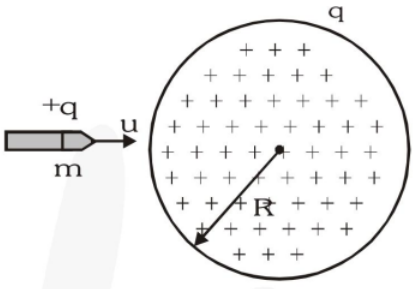
(A) $W_2 + W_1 - K_1$

(B) $K_1 - W_2 - W_1$

(C) $W_1 - W_2 + K_2 - K_1$

(D) $W_2 - W_1 + K_1 - K_2$

**Q12.** A bullet of mass m and charge q is fired towards a solid uniformly charged sphere of radius R and total charge + q. If it strikes the surface of sphere with speed u, find the minimum value of u so that it can penetrate through the sphere. (Neglect all resistnace forces or friction acting on bullet except electrostatic forces)




(A) $\frac{q}{\sqrt{2\pi\varepsilon_0 mR}}$ (B) $\frac{q}{\sqrt{4\pi\varepsilon_0 mR}}$

(C) $\frac{q}{\sqrt{8\pi\varepsilon_0 mR}}$ (D) $\frac{\sqrt{3}q}{\sqrt{4\pi\varepsilon_0 mR}}$

**Q13.** The electrostatic potential on the surface of a charged conducting solid sphere is 100V. Two statements are made in this regard:

$S_1$ : At any point inside the sphere, electric intensity is zero.

$S_2$ : At any point inside the sphere, the electrostatic potential is 100V.

Which of the following is a correct statement?

(A) $S_1$ is true but $S_2$ is false.

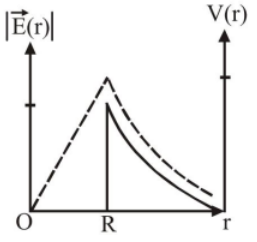
(B) Both $S_1$ & $S_2$ are false.

(C) $S_1$ is true, $S_2$ is also true and $S_1$ is the cause of $S_2$.

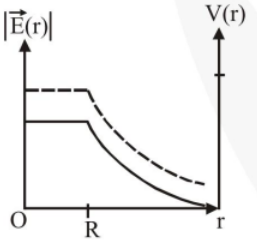
(D) $S_1$ is true, $S_2$ is also true but the statements are independent.

**Q14.** Consider a thin conducting spherical shell of radius R with its centre at the origin, carrying uniform positive surface charge density. The variation of the magnitude of the electric field $|\vec{E}(r)|$ and the electric potential V(r) with the distance r from the centre, is best represented by which graph:

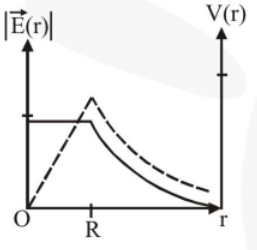
(A)



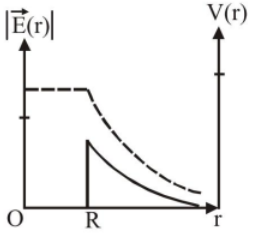

(B)



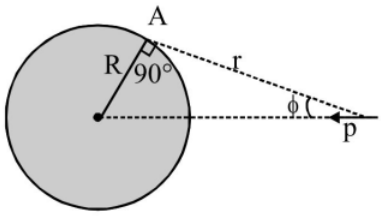

(C)




(D)




**Q15.** A dipole having dipole moment p is placed in front of a solid uncharged conducting sphere as shown in the diagram. The net potential at point A lying on the surface of the sphere is




(A) $\frac{kp\cos\phi}{r^2}$ (B) $\frac{kp\cos^2\phi}{r^2}$

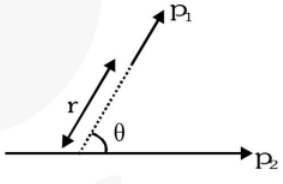
(C) $\frac{2kp\cos^2\phi}{r^2}$ (D) zero

**Q16.** A small electric dipole is of dipole moment p. The electric potential at a distance 'r' from its centre and making an angle θ from the axis of dipole will be:-

(A) $\frac{kp\sin\theta}{r^2}$ (B) $\frac{kp\cos\theta}{r^2}$

(C) $\frac{kp}{r^3}\sqrt{1+3\cos^2\theta}$ (D) $\frac{kp}{r^3}\sqrt{1+3\sin^2\theta}$

**Q17.** Two short electric dipoles are placed as shown. The energy of electric interaction between these dipoles will be :




(A) $\frac{2Kp_1p_2\cos\theta}{r^3}$ (B) $\frac{-2Kp_1p_2\cos\theta}{r^3}$

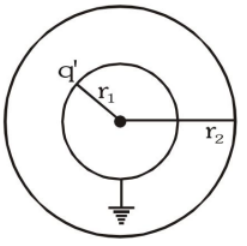
(C) $\frac{-2Kp_1p_2\sin\theta}{r^3}$ (D) $\frac{-4Kp_1p_2\cos\theta}{r^3}$

**Q18.** If n drops of equal size and same potential V are converted into a bigger drop then the potential of bigger drop will be :–

(A) $\frac{V}{n}$ (B) Vn

(C) $Vn^{1/3}$ (D) $Vn^{2/3}$

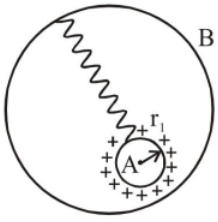
**Q19.** The concentric conducting spheres are of radii $r_1$ and $r_2$. The outer sphere is given a charge q. The charge q′ on the inner sphere will be (inner sphere is grounded):




(A) q (B) -q

(C) $-q\frac{r_1}{r_2}$ (D) Zero

**Q20.** A metal sphere A of radius $r_1$ charged to a potential $\phi_1$ is enveloped by a thin walled conducting spherical shell B of radius $r_2$. Then $\phi_2$ of the sphere A after it is connected by a thin conducting wire to the shell B will be:




(A) $\phi_1\left(\frac{r_1}{r_2}\right)$ (B) $\phi_1\left(\frac{r_2}{r_1}\right)$

(C) $\phi_1\left(1-\frac{r_2}{r_1}\right)$ (D) $\phi_1\left(\frac{r_1 r_2}{r_1+r_2}\right)$

## *Exercise - 2A Brain Booster*

### ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** A charged particle having some mass is resting in equilibrium at a height H above the centre of a uniformly charged non-conducting horizontal ring of radius R. The force of gravity acts downwards. The equilibrium of the particle will be stable.

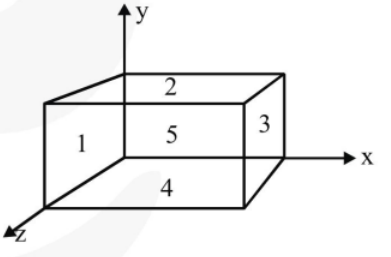
(A) for all values of H

(B) only if $H > \frac{R}{\sqrt{2}}$

(C) only if $H < \frac{R}{\sqrt{2}}$

(D) only if $H = \frac{R}{\sqrt{2}}$

**Q2.** In a region of space, the electric field $\vec{E} = E_0 x\hat{i} + E_0 y\hat{j}$. Consider an imaginary cubical volume of edge 'a' with its edges parallel to the axes of coordinates. Now,



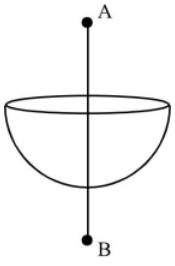

(A) the total electric flux through the faces 1 and 3 is $E_0a^3$

(B) the charge inside the cubical volume is $2\varepsilon_0E_0a^3$

(C) the total electric flux through the faces 2 and 4 is $2E_0a^3$

(D) the charge inside the cubical volume is $\varepsilon_0E_0a^3$

**Q3.** The diagram shows a uniformly charged hemisphere of radius R. It has volume charge density ρ. If the electric field at a point 2R distance above its center is E then what is the electric field at the point which is 2R below its center?




(A) $\rho R/6\varepsilon_0 + E$ (B) $\rho R/12\varepsilon_0 - E$
(C) $-\rho R/6\varepsilon_0 + E$ (D) $\rho R/24\varepsilon_0 + E$

**Q4.** Mark the correct option(s) :

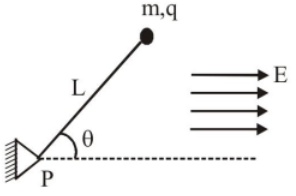
(A) Gaussian surface can pass through discrete charge.

(B) Gaussian surface can be of any shape.

(C) The electric field calculated by Gauss's law is the field only due to the charges inside the Gaussian surface.

(D) The flux of the electric field through a closed surface due to all the charges is equal to the flux due to the charges enclosed by the surface.

**Q5.** A particle of mass m and charge q is attached to a light rod of length L. The rod can rotate freely in the plane of paper about the other end, which is hinged at P. The entire assembly lies in a uniform electric field E also acting in the plane of paper as shown. The rod is released from rest when it makes an angle θ with the electric field direction. Determine the speed of the particle when the rod is parallel to the electric field.




(A) $\left(\frac{2qEL(1-\cos\theta)}{m}\right)^{1/2}$

(B) $\left(\frac{2qEL(1-\sin\theta)}{m}\right)^{1/2}$

(C) $\left(\frac{qEL(1-\cos\theta)}{2m}\right)^{1/2}$

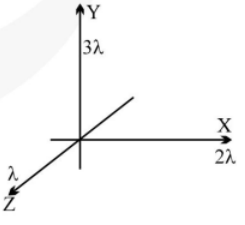
(D) $\left(\frac{2qEL\cos\theta}{m}\right)^{1/2}$

**Q6.** A charged particle of charge Q is held fixed and another charged particle of mass m and charge q (of the same sign) is released from a distance r. The impulse of the force exerted by the external agent on the fixed charge by the time distance between Q and q becomes 2r is -

(A) $\sqrt{\frac{Qq}{4\pi\epsilon_0 mr}}$ (B) $\sqrt{\frac{Qqm}{4\pi\epsilon_0 r}}$

(C) $\sqrt{\frac{Qqm}{\pi\epsilon_0 r}}$ (D) $\sqrt{\frac{Qqm}{2\pi\epsilon_0 r}}$

**Q7.** The diagram shows three infinitely long uniform line charges placed on the X, Y and Z axis. The work done in moving a unit positive charge from (1, 1, 1) to (0, 1, 1) is equal to -




(A) $\frac{\lambda \ln 2}{2\pi\varepsilon_o}$ (B) $\frac{\lambda \ln 2}{\pi\varepsilon_o}$

(C) $\frac{3\lambda \ln 2}{2\pi\varepsilon_o}$ (D) None

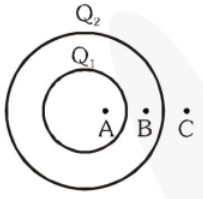
**Q8.** In an electric field the potential at a point is given by the following relation $V = \frac{343}{r}$.

The electric field at $\vec{r} = 3\hat{i} + 2\hat{j} + 6\hat{k}$ is :

(A) $21\hat{i} + 14\hat{j} + 42\hat{k}$ (B) $3\hat{i} + 2\hat{j} + 6\hat{k}$

(C) $\frac{1}{7}\left(3\hat{i} + 2\hat{j} + 6\hat{k}\right)$ (D) $-\left(3\hat{i} + 2\hat{j} + 6\hat{k}\right)$

**Q9.** Figure shows two uniform charged concentric spherical shell. Both charges are positive, Select correct statement :



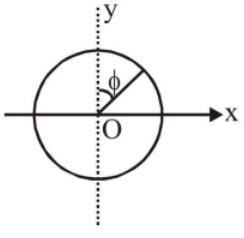

(A) Electric field intensity at B may be greater than electric field intensity at C.

(B) Electric field intensity at B must be greater than electric field intensity at C.

(C) Potential at A greater than potential at B

(D) If a charge moves from B to C work done by electric force must be positive.

**Q10.** A thin non-conducting ring of radius a has a linear charge density $\lambda = \lambda_0 \sin\phi$. A uniform electric field $E_0\hat{i} + E_0\hat{j}$ is present there. Net torque acting on ring is -




(A) $E_0\sqrt{2}\pi a^2\lambda_0$ (B) $E_0\pi a^2\lambda_0$

(C) $2E_0\pi a^2\lambda_0$ (D) zero

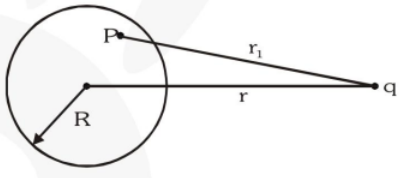
**Q11.** Two non-conducting hemispherical surfaces, which are having uniform charge density $\sigma$ are placed on smooth horizontal surface as shown in figure. Assuming springs are ideal, calculate compression in each spring if both the hemispherical surface are just touching each other.




(A) $\frac{\sigma^2 R^2}{2\varepsilon_0 k}$ (B) R

(C) $\frac{\sigma^2 \pi R^2}{2\varepsilon_0 k}$ (D) None of these

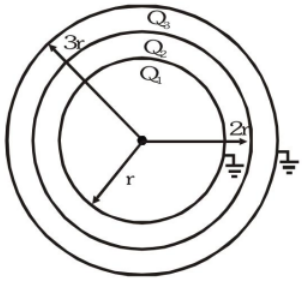
**Q12.** A point charge q is placed at a distance r from center of a conducting neutral sphere of radius R (r>R). The potential at any point P inside the sphere at a distance $r_1$ from point charge due to induced charge of the sphere is given by-




(A) $\frac{Kq}{r_1} - \frac{Kq}{R}$ (B) $\frac{Kq}{r_1} - \frac{Kq}{R}$

(C) $\frac{Kq}{r} - \frac{Kq}{r_1}$ (D) $-\frac{Kq}{r_1} + \frac{Kq}{R}$

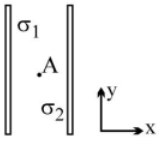
**Q13.** Three concentric conducting spherical shells have radius r, 2r and 3r and $Q_1$, $Q_2$ and $Q_3$ are final charges respectively. Innermost and outermost shells are already earthed as shown in figure. choose the wrong statement.



(A) $Q_1 + Q_3 = -Q_2$ (B) $Q_1 = \frac{-Q_2}{4}$

(C) $\frac{Q_3}{Q_1} = 3$ (D) $\frac{Q_3}{Q_2} = \frac{-1}{3}$

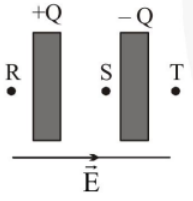
**Q14.** Two large conducting sheets are kept parallel to each other as shown. In equilibrium, the charge density on facing surfaces is $\sigma_1$ and $\sigma_2$. What is the value of electric field at A.




(A) $\frac{\sigma_1}{\varepsilon_0}\hat{i}$ (B) $-\frac{\sigma_2}{\varepsilon_0}\hat{i}$

(C) $\frac{\sigma_1 + \sigma_2}{2\varepsilon_0}\hat{i}$ (D) $\frac{\sigma_1 - \sigma_2}{2\varepsilon_0}\hat{i}$

**Q15.** Two large thin conducting plates with small gap in between are placed in a uniform electric field E (perpendicular to the plates). Area of each plate is A and charges +Q and –Q are given to these plates as shown in the figure. If points R,S and T as shown in the figure are three points in space, then the




(A) Field at point $R$ is $E$

(B) Field at point $S$ is $E$

(C) Field at point $T$ is $\left(E + \frac{Q}{\varepsilon_o A}\right)$

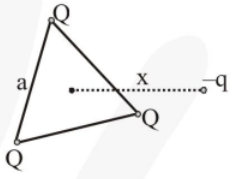
(D) Field at point $S$ is $\left(E + \frac{Q}{A\varepsilon_o}\right)$

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph for Q. No. 16 to 18**

Three charged particles each of +Q are fixed at the corners of an equilateral triangle of side 'a'. A fourth particle of charge –q and mass m is placed at a point on the line passing through centroid of triangle and perpendicular to the plane of triangle at a distance x from the centre of triangle.




**Q16.** Magnitude of resultant force on the fourth charged particle is

(A) $\frac{1}{4\pi \in_0} \frac{9\sqrt{3}\ Qqx}{(3x^2 + a^2)^{3/2}}$

(B) $\frac{1}{4\pi \in_0} \frac{27\sqrt{3}\ Qqx}{(3x^2 + a^2)^{3/2}}$

(C) $\frac{1}{4\pi \in_0} \frac{2\sqrt{2}\ Qqx}{(2x^2 + a^2)^{3/2}}$

(D) $\frac{1}{4\pi \in_0} \frac{4\sqrt{2}\ Qqx}{(2x^2 + a^2)^{3/2}}$

**Q17.** Value of x for which the force is maximum is

(A) $\frac{a}{\sqrt{3}}$ (B) $\frac{a}{\sqrt{2}}$

(C) $\frac{a}{\sqrt{6}}$ (D) $\frac{a}{\sqrt{5}}$

**Q18.** For small oscillation the period of oscillation of fourth particle is

(A) $2\pi\sqrt{\frac{4\pi\epsilon_0 ma^3}{9\sqrt{3}Qq}}$

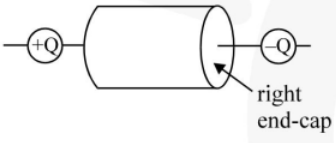
(B) $\pi\sqrt{\frac{4\pi\epsilon_0 ma^3}{9\sqrt{3}Qq}}$

(C) $2\pi\sqrt{\frac{2\pi\epsilon_0 ma^3}{27\sqrt{3}Qq}}$

(D) $2\pi\sqrt{\frac{\pi\epsilon_0 ma^3}{27\sqrt{3}Qq}}$

**Paragraph for Question No. 19 & 20**

The figure applies to the following two questions. Positive and negative charges of equal magnitude lie along the symmetry axis of a cylinder. The distance from the positive charge to the left end-cap of the cylinder is the same as the distance from the negative charge to the right end -cap.




**Q19.** What is the flux of the electric field through the closed cylinder ?

(A) 0 (B) $+Q/\varepsilon_0$

(C) $+2Q/\varepsilon_0$ (D) $-Q/\varepsilon_0$

**Q20.** What is the sign of the flux through the right end-cap of the cylinder ?

(A) Positive

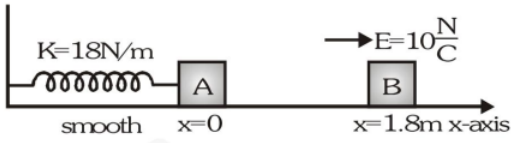
(B) Negative

(C) There is no flux through the right end-cap.

(D) None of these

**Paragraph for Question No. 21 to 23**

Electrostatic force on a charged particle is given by $\vec{F} = q\vec{E}$. If q is positive $\vec{F} \uparrow\uparrow \vec{E}$ and if q negative $\vec{F} \uparrow\downarrow \vec{E}$. In the figure $m_A = m_B = 1kg$. Block A is neutral while $q_B = -1C$. Sizes of A and B are negligible. B is released from rest at a distance 1.8 m from A. Initially spring is neither compressed nor elongated.




**Q21.** If collision between A and B is perfectly inelastic, what is velocity of combined mass just after collision ?

(A) 6m/s (B) 3m/s

(C) 9m/s (D) 12m/s

**Q22.** Equilibrium position of the combined mass is at x = .....m.

(A) $-\frac{2}{9}$ (B) $-\frac{1}{3}$

(C) $-\frac{5}{9}$ (D) $-\frac{7}{9}$

**Q23.** The amplitude of oscillation of the combined mass will be :–

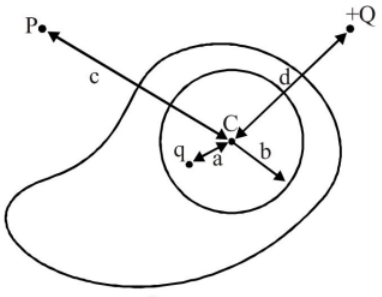
(A) $\frac{2}{3}$m (B) $\frac{\sqrt{124}}{3}$m

(C) $\frac{\sqrt{72}}{9}$m (D) $\frac{\sqrt{106}}{9}$m

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q24.** In the shown figure the conductor is uncharged and a charge $q$ is placed inside a spherical cavity at a distance $a$ from its centre ($C$). Point $P$ and charge $+Q$ are as shown. $a$, $b$, $c$, $d$ are known.



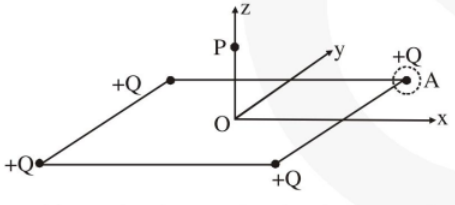

| **Column-I** | **Column-II** |
|---|---|
| (A) Electric field due to induced charges on the inner surface of cavity at point $P$ | (P) zero |
| (B) Electric potential due to charges on the inner surface of cavity and $q$ at $P$ | (Q) non-zero |
| (C) Electric field due to induced charges on the outer surface of conductor and $Q$ at $C$ | (R) value can be stated with the given data. |
| (D) Electric potential due to induced charges on the inner surface of cavity at $C$ | (S) value cannot be stated from the given data. |

**Q25.** Column I shows certain charge configurations in x-y plane. Resultant field is $\vec{E} = E_x\hat{i} + E_y\hat{j} + E_2\hat{k}$.

If a component points along positive x-axis it is conventionally positive i.e. $E_x > 0$ implies x component is along positive x-axis. Point O is equidistant from all charges.

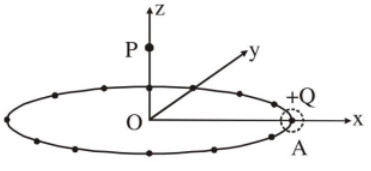
**Column-I**

(A)




Four point charges placed at the vertex of a square

(B)




N (even number more than 6) identical point charges are placed symmetrically on a non-conducting ring.

**Column-II**

(P) Resultant field at P has

$E_x = 0$

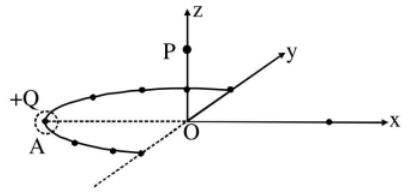
$E_y = 0$

$E_z \neq 0$

(Q) Resultant field at O has

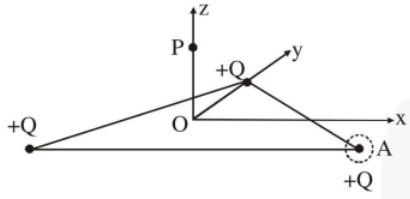
$E_x = E_y = E_z = 0$

(C)




N (is odd number more than 5) identical point charges are placed symmetrically on a non-conducting semi-circular ring.

(D)




Three charges are placed on the vertex of an equilateral triangle. Point O is centroid of triangle.

(R) If charge at A is removed, resultant field at P has
$E_z > 0$
$E_x > 0$
$E_y > 0$

(S) If charged at A is removed; resultant field at O has $E_x > 0$

(T) If charge at A becomes negative, resultant field at O
$E_x > 0$
$E_y > 0$

# Answer Key

| Ex-1 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** |
| D | A | B | **D** | B | A | B | C | C | B |
| **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** | **19** | **20** |
| A | D | C | A | D | B | **B** | B | A | B |
| **21** | **22** | **23** | **24** | **25** | **26** | **27** | **28** | **29** | **30** |
| B | C | B | B | C | C | A | D | C | A |
| **31** | **32** | **33** | **34** | **35** | **36** | **37** | **38** | **39** | |
| C | **D** | **C** | **A** | **D** | **D** | **C** | **B** | A | |

## Ex-1A

**1.** A,C **2.** A,C,D **3.** B,C **4.** A,B,C,D **5.** A, B

**6.** A,B,D **7.** A **8.** A,B,C **9.** A,C **10.** B,C

**11.** A,D **12.** B **13.** B **14.** A,B,C **15.** D

**16.** A,D **17.** D **18.** A **19.** B **20.** D

**21.** A $\rightarrow$ (P,R) ; B $\rightarrow$ (P,Q,S); C $\rightarrow$ (P,Q); D $\rightarrow$ (P,Q)

**22.** $15\text{Nm}^2/\text{C}$ **23.** $11.38\times10^{-19}\text{kg}$

**24.** $\frac{\lambda}{4\pi\epsilon_0 R}$ **25.** $q = -2\sqrt{2}Q$

| Ex-2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** |
| D | D | D | C | A | C | D | A | B | B |
| **11** | **12** | **13** | **14** | **15** | **16** | **17** | **18** | **19** | **20** |
| B | B | C | D | B | B | B | D | C | A |

## Ex-2A

**1.** B **2.** A,B **3.** B **4.** B,D **5.** A

**6.** B **7.** B **8.** B **9.** A,C **10.** B

**11.** C **12.** C **13.** D **14.** A,B,D **15.** A,D

**16.** A **17.** C **18.** A **19.** A **20.** A

**21.** B **22.** C **23.** D

**24.** A $\rightarrow$ (Q,S) ; B $\rightarrow$ (P,R); C $\rightarrow$ (P,R); D $\rightarrow$ (Q,R)

**25.** A $\rightarrow$ (P,Q,R,S,T) ; B $\rightarrow$ (P,Q,S); C $\rightarrow$ (S); D $\rightarrow$ (P,Q,S)

## -: Important Notes & Formulas :-